

BatchRunControl – A Script for HEC-ResSim

General

The BatchRunControl script (BatchRunControl.py) provides a graphical interface that allows users to select multiple pre-defined HEC-ResSim simulation computations from a single watershed and execute them in a batch mode.

Execution

The script is executed by giving its filename as the first parameter to the HEC-ResSim command. The directory of the desired watershed is given as the first parameter to the script (second parameter to the HEC-ResSim command) as follows:

```
HEC-ResSim c:\scripts\BatchRunControl.py c:\watershed\base\MyWatershed
```

User Interface

The user interface consists of two windows: a Log Window and a Main Window. The Log Window consists of a scrolling text area, a progress bar, and a status bar. The Log Window displays current HEC-ResSim output regardless of which log file is currently in use. The Log Window appears when the script first starts and displays the progress of reading the simulations from the specified watershed (Figure 1). The Main Window is displayed after all the simulations have been read.

The Main Window consists of a tree pane and a button pane (Figure 2). The tree pane contains a tree with a folder for each simulation in the watershed. Each simulation folder contains nodes for each alternative available with that simulation. The alternative nodes are leaf nodes if no trials are defined. If trials are defined for an alternative, the alternative node is a folder containing leaf nodes for each trial plus one leaf node for the base alternative.

Folder nodes can be expanded or collapsed by clicking on the  or  icon next to the node or by double-clicking on the folder. All folder nodes can be expanded by clicking the **Expand All** button or collapsed by clicking the **Collapse All** button.

Individual computations can be selected or deselected by clicking the checkbox next to its node. Clicking the checkbox next to a folder node selects or deselects all nodes in the folder. All nodes in the tree can be selected or deselected by clicking the checkbox next to the root (Simulations) folder or by clicking the **Select All** or **Select None** buttons. Clicking the **Invert Selections** button reverses the state of all current selections.

By default only simulation names are displayed at their folder nodes. To see the lookback, start and end times for the simulations, check the **Show Times** checkbox in the button pane (Figure 3).

Clicking the **Cancel** button exits the script, even if it is currently running computations. Clicking the **Run Selected** button starts running the computations for each leaf node selected. During this time the Log Window displays the HEC-ResSim output and indicates the current progress and status (Figure 4).

Log Files

Log files are written to the simulation directory specified on the command line and are named `BatchRunControl_YYYY-MM-DD_hhmm_id.log`, where `YYYY-MM-DD_hhmm` is the date and time the log file was opened, and *id* is a descriptive identifier. The `Startup` identifier indicates that the log file contains output generated while the watershed was being read. The `Simulations` identifier indicates that the log file contains output generated while running computations. The simulations log file contains output from the script indicating when it started specific computations, as shown below:

```
2010/01/27-22:58:52 ###  
2010/01/27-22:58:52 ### Computing simulation 2006, alternative FC-GageQ, trial Try1  
2010/01/27-22:58:52 ###
```

If the **Separate Log Files** checkbox is checked in the Main Window, the simulations log file will contain only the script output as shown above plus a small amount of HEC-ResSim output that occurs between simulations. The HEC-ResSim output that occurs during simulations will be contained in log files with identifiers like *Simulation{Alternative}* or *Simulation{Alternative{Trial}}* (Figure 5).

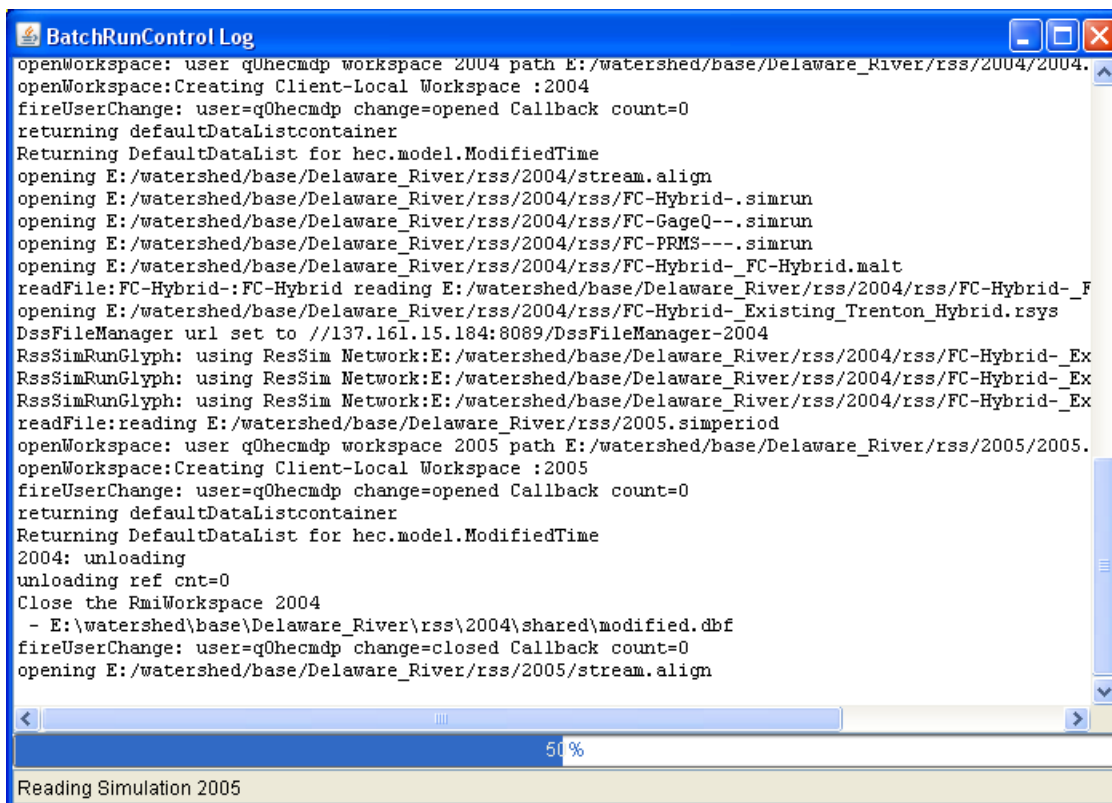


Figure 1. Log Window During Startup

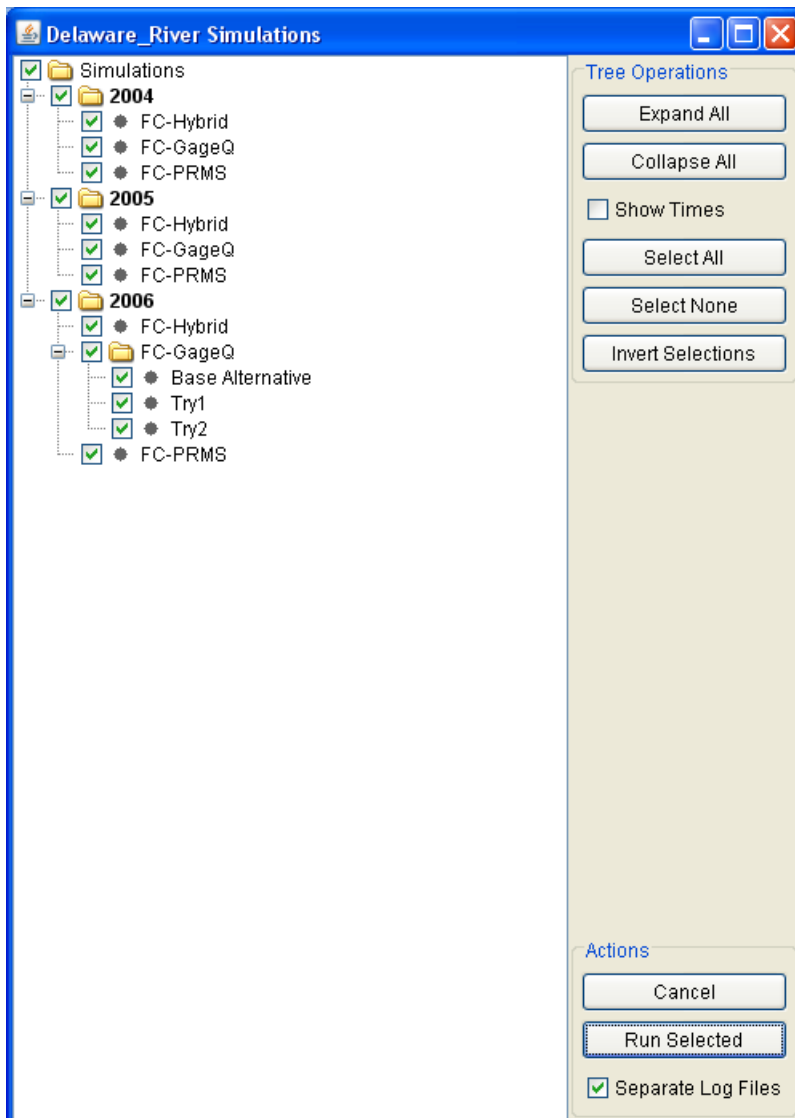


Figure 2. Main Window

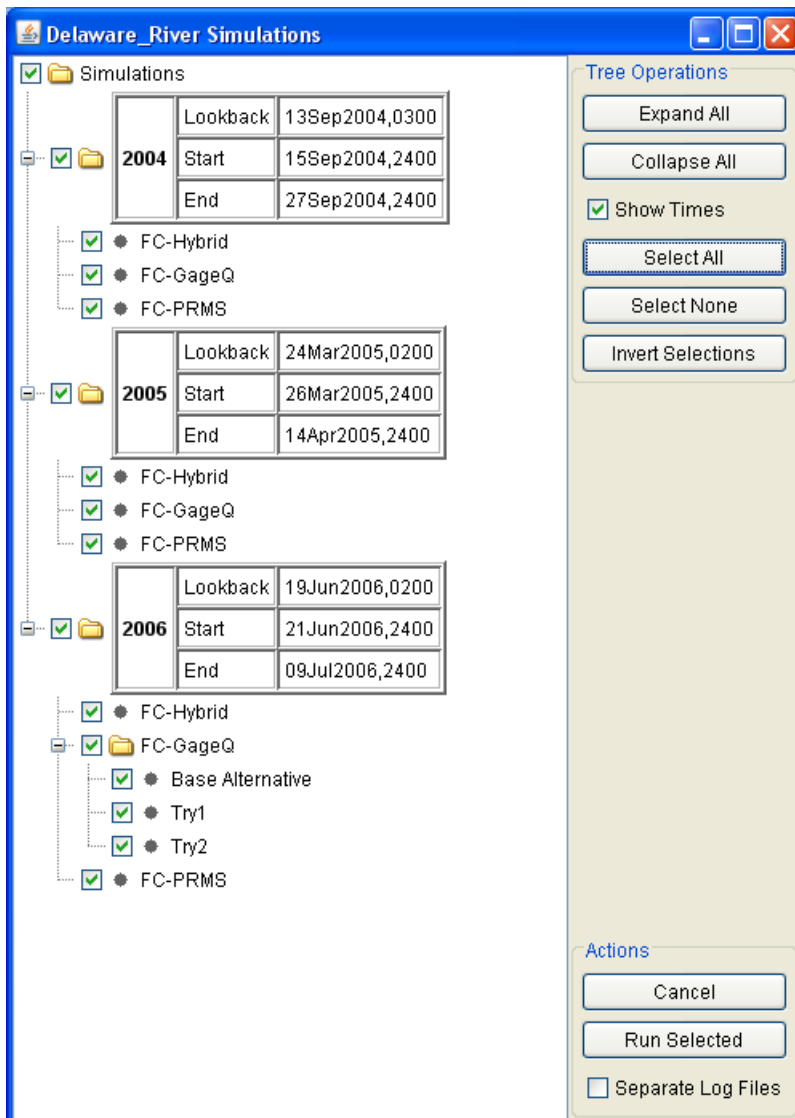


Figure 3. Main Window with **Show Times** Selected

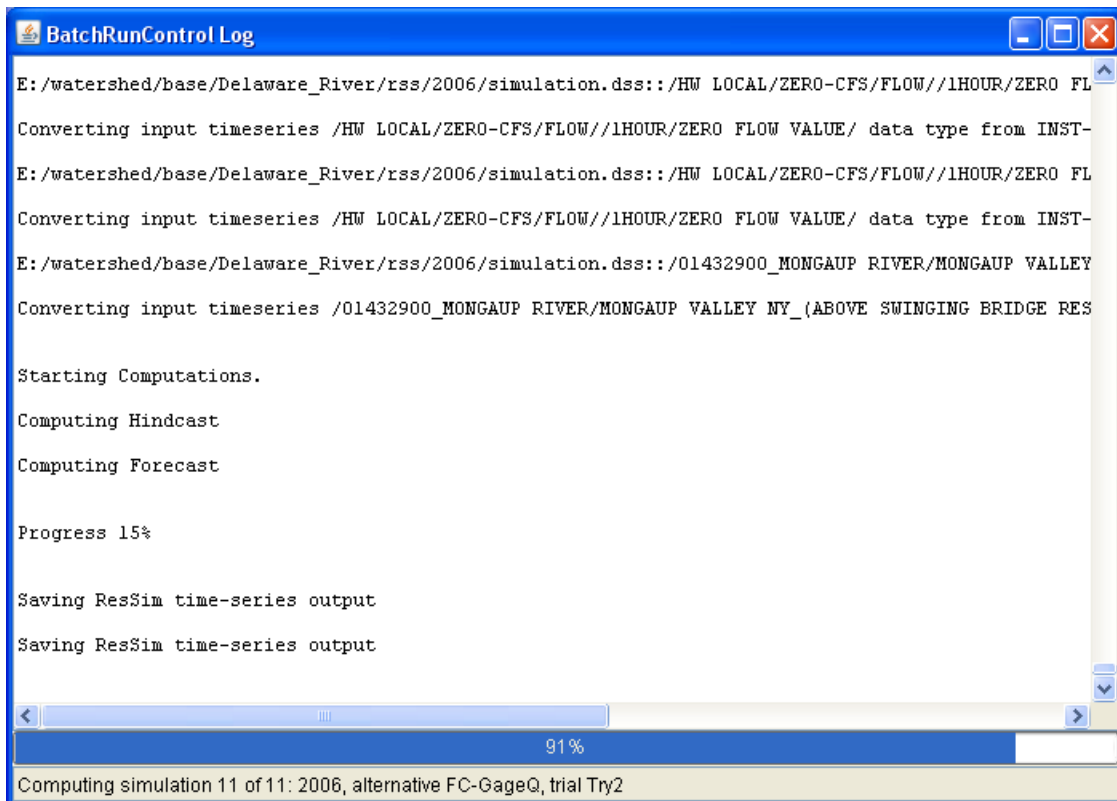


Figure 4. Log Window During Computations

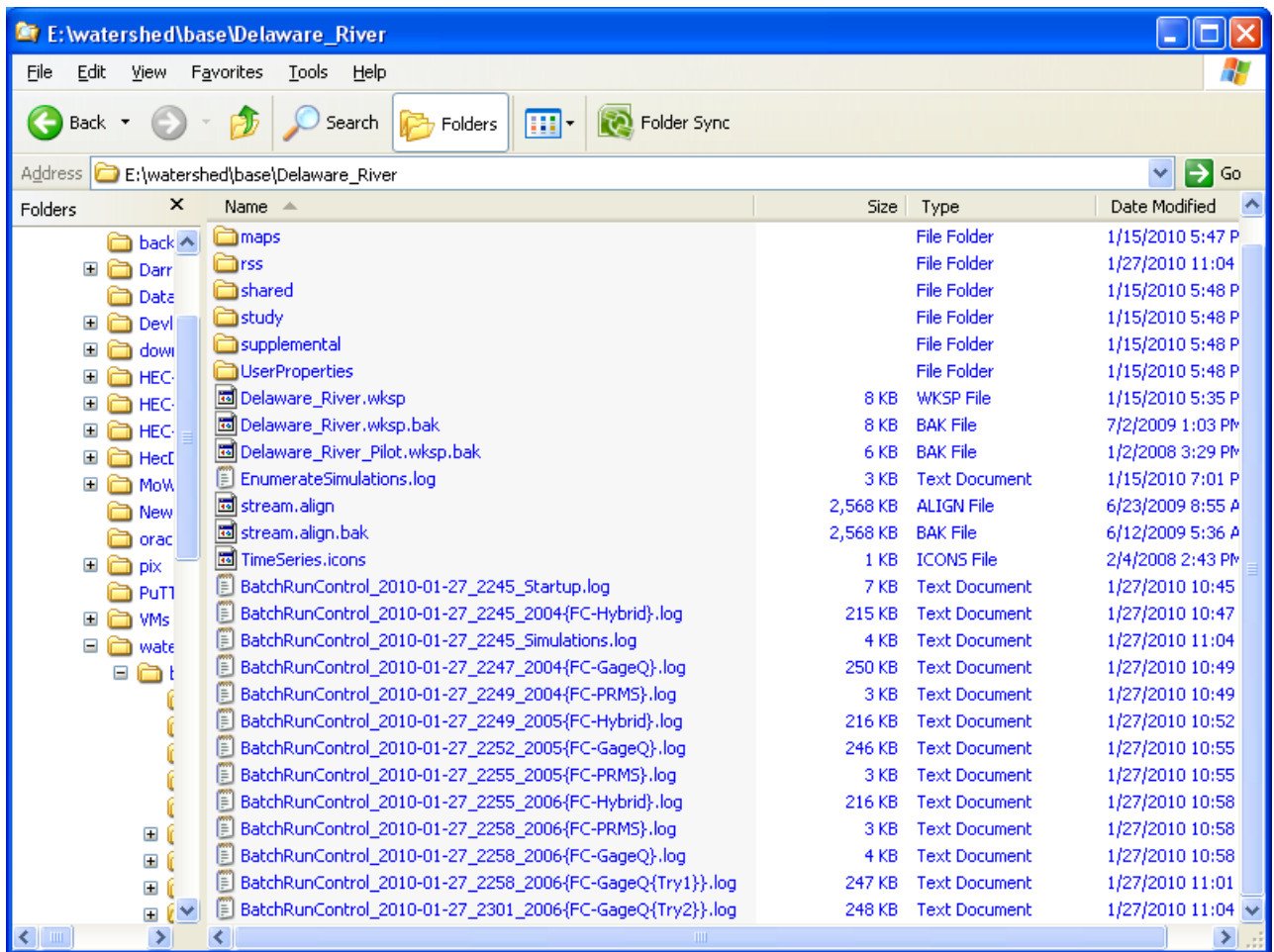


Figure 5. Log File Location and Names